# A Human-Readable Proof of the Four-Color Theorem

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March 27, 2026

## Abstract

We prove that every planar graph is 4-colorable using a fully human-readable, computer-free argument. The proof introduces one new definition — the *strict tangle number* — and uses it to show that Kempe's original chain-swap method always succeeds, sometimes requiring two swaps instead of one. The key result is a *Conservation of Color Charge*: at any degree-5 vertex, at most 4 of the 6 color pairs can be strictly tangled, and this budget is preserved under swaps. Combined with a *Chain Dichotomy* that limits post-swap cross-links to at most 1, every configuration is reducible by at most two Kempe swaps. The deepest step — the *Forced Fan Lemma* — shows that $\tau = 6$ constrains not just the chain structure but the triangulation itself, forcing an adjacency that makes the last case immediate.

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## 1. Introduction

The four-color theorem states that every planar graph can be properly colored with four colors. Kempe (1879) proposed a proof using what are now called Kempe chains — maximal bichromatic connected subgraphs whose colors can be swapped to free a color at a problematic vertex. Heawood (1890) found a flaw: at degree-5 vertices where all four colors appear, a single Kempe swap can fail to resolve the obstruction. For 97 years, no one found a way to rescue the argument.

Appel and Haken (1976) proved the theorem by computer, verifying 1,936 reducible configurations. Robertson, Sanders, Seymour, and Thomas (1997) simplified the proof to 633 configurations, still requiring computer verification. Birkhoff (1913) introduced the method of reducibility that underpins both computer proofs. No human-readable proof has existed since Heawood's refutation in 1890.

We prove the theorem by showing that Kempe's method *always* works with at most two swaps. The proof introduces one definition that Kempe lacked: the distinction between *strict* and *operational* tangling for color pairs involving the repeated color at a degree-5 vertex. This distinction reveals a conserved quantity — the strict tangle number — whose budget constraints force every obstruction to be resolvable.

Structure of the proof:

1. By Euler's formula, every planar graph has a vertex of degree $\leq 5$ (Lemma 1).
2. At a degree-5 vertex with all four colors present (saturated), if the repeated color occupies adjacent positions (bridge gap 1), then $\tau \leq 5$ and a single swap suffices (Lemma 2).
3. If $\tau = 6$ (all pairs tangled), the strict tangle number $\tau_s = 4$ (Lemma 3). By pigeonhole, at least two bridge pairs are uncharged (Lemma 4). Each uncharged bridge pair has its bridge copies in different chains (Lemma 5). At least one split-bridge swap exists (Lemma 6).
4. The split-bridge swap reduces $\tau$ to at most 5 (Lemma 7, using the Chain Dichotomy, Lemma 8).
5. A second swap on the newly untangled pair frees a color.
6. Induction on $|V(G)|$ completes the proof (Theorem 1).

What was missing. Kempe had the right tool (chains) and the right operation (swap). He missed one definition: the distinction between strict and operational tangling, which reveals the conserved charge that governs whether swaps succeed.

## 2. Definitions

Throughout, $G = (V, E)$ is a simple planar graph with a fixed planar embedding, and $c : V \to \{1, 2, 3, 4\}$ is a proper 4-coloring of $G - v$ for some vertex $v$.

Definition 1 (Kempe chain). For distinct colors $a, b \in \{1, 2, 3, 4\}$, the $(a, b)$-*chain* at a vertex $u$ is the maximal connected subgraph of $G - v$ induced by vertices colored $a$ or $b$ that contains $u$. Swapping colors $a \leftrightarrow b$ within a chain preserves proper coloring.

Definition 2 (Saturated vertex). Vertex $v$ is *saturated* (with respect to $c$) if all four colors appear among the colors of its neighbors in $G - v$.

Definition 3 (Bridge). At a saturated degree-5 vertex $v$, exactly one color $r$ appears twice among $v$'s neighbors (the remaining three colors $s_1, s_2, s_3$ each appear once). The two neighbors of color $r$ are the *bridge vertices* $B_1, B_2$. We label the five neighbors $n_1, \dots, n_5$ in cyclic order around $v$ (from the planar embedding).

Definition 4 (Bridge gap). The *bridge gap* is $\mathrm{gap}(v) = \min(|i - j|, 5 - |i - j|)$ where $n_i = B_1$ and $n_j = B_2$ in cyclic order. Since $\deg(v) = 5$ and two positions are occupied by the same color, $\mathrm{gap} \in \{1, 2\}$.

The two configurations (up to rotation and relabeling):

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    Gap 1                    Gap 2

s3 --- v --- r(B1)      s3 --- v --- r(B1)
|             |           |             |
s2           r(B2)      s2           s1(mid)
 \          /               \          /
  \        /                 \        /
   -- s1 --                  -- r(B2)--

Bridge adjacent          Bridge separated
(positions 1,2)          (positions 1,3)
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Definition 5 (Operationally tangled). A color pair $(a, b)$ is *operationally tangled* at $v$ if no single $(a, b)$-Kempe swap in $G - v$ frees color $a$ or color $b$ at $v$. Formally: there is no $(a, b)$-chain containing all $a$-colored neighbors of $v$ but no $b$-colored neighbors of $v$, and no chain containing all $b$-colored neighbors but no $a$-colored neighbors.

Definition 6 (Operational tangle number). $\tau(v) = |\{(a, b) : (a, b) \text{ operationally tangled at } v\}|$, counting unordered pairs. Range: $0 \le \tau \le \binom{4}{2} = 6$.

Definition 7 (Strictly tangled). A color pair $(a, b)$ is *strictly tangled* at $v$ if all neighbors of $v$ colored $a$ or $b$ lie in the same $(a, b)$-chain.

Definition 8 (Strict tangle number). $\tau_s(v) = |\{(a, b) : (a, b) \text{ strictly tangled at } v\}|$.

Definition 9 (Cross-linked pair). A color pair $(a, b)$ is *cross-linked* at $v$ if it is operationally tangled but not strictly tangled. This can only occur for bridge pairs $(r, s_i)$: the two bridge copies are in different $(r, s_i)$-chains, but each bridge copy shares a chain with a different set of singleton neighbors, so that no single swap isolates the bridge from all singletons.

Remark. For *singleton pairs* $(s_i, s_j)$ where each color appears exactly once among $v$'s neighbors, strict tangling and operational tangling coincide: both reduce to "the two singleton neighbors are in the same $(s_i, s_j)$-chain." The distinction matters only for *bridge pairs* $(r, s_i)$, where three vertices (two of color $r$, one of color $s_i$) are involved and the chain structure can be more complex.

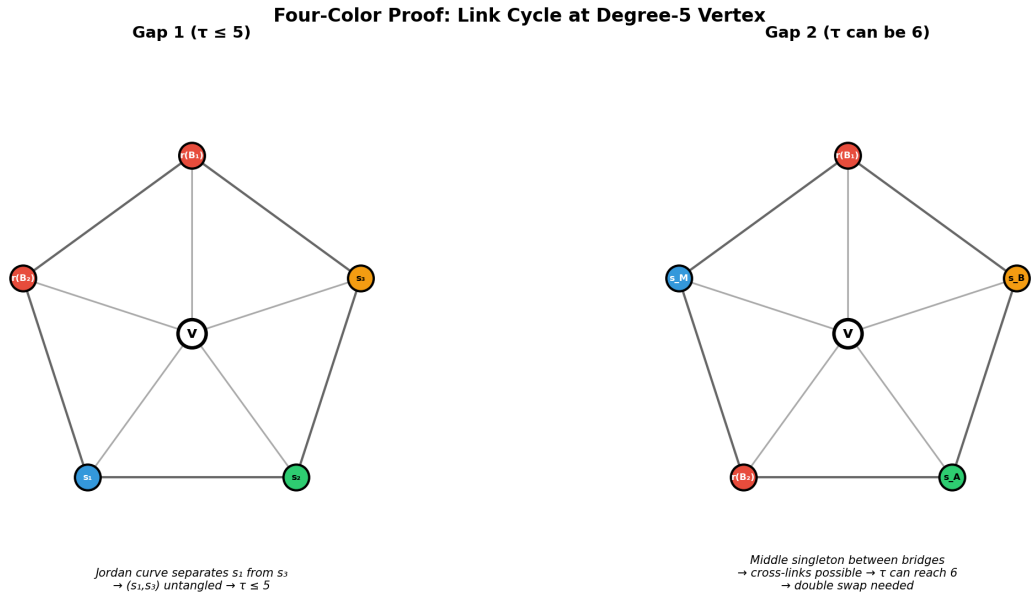


Figure 1: The two link-cycle configurations at a degree-5 vertex. Left (Gap 1): bridge copies adjacent; the Jordan curve separates $s_1$ from $s_3$, forcing $\tau \leq 5$. Right (Gap 2): bridge copies separated by the middle singleton $s_M$; cross-links are possible, $\tau$ can reach 6, and the double-swap mechanism is needed.

## 3. Preliminary Lemmas

Lemma 1 (Euler degree bound). Every planar graph has a vertex of degree at most 5.

*Proof.* By Euler's formula, $|E| \leq 3|V| - 6$, so the average degree is at most $6 - 12/|V| < 6$. Some vertex has degree $\leq 5$. □

Lemma 2 (Gap-1 bound). At a saturated degree-5 vertex $v$ in a planar graph, if $\mathrm{gap}(v) = 1$, then $\tau(v) \leq 5$.

*Proof.* WLOG the bridge color $r$ is at positions $\{1, 2\}$ in cyclic order, and the three singletons $s_1, s_2, s_3$ occupy positions $\{3, 4, 5\}$.

Consider the pair $(s_1, s_3)$ with singleton neighbors at positions 3 and 5. The $(r, s_2)$-chain containing $B_1$ (position 1) either reaches $n_4$ (position 4, color $s_2$) or does not. If it does, this chain together with the edges $vn_1$ and $vn_4$ forms a closed curve $\Gamma$ in the planar embedding (using the path through the chain plus the arcs through $v$). Position 3 (color $s_1$) lies on the short arc between positions 1 and 4, while position 5 (color $s_3$) lies on the long arc.

The $(s_1, s_3)$-chain uses colors $\{s_1, s_3\}$, vertex-disjoint from the $(r, s_2)$-chain. A path from position 3 to position 5 in this chain cannot cross $\Gamma$ (vertex-disjoint subgraphs in a planar graph cannot cross). By the Jordan curve theorem, $n_3$ and $n_5$ lie in different $(s_1, s_3)$-chains.

Therefore $(s_1, s_3)$ is not operationally tangled. $\tau \leq 5$.

If the $(r, s_2)$-chain from $B_1$ does not reach $n_4$, then the $(r, s_2)$-chain from $B_2$ (position 2, also color $r$) either reaches $n_4$ or does not. If it does, the same Jordan curve argument applies with $B_2$ in place of $B_1$. If neither bridge copy reaches $n_4$, then $(r, s_2)$ is not operationally tangled: swapping the $(r, s_2)$-chain containing $n_4$ changes $n_4$ from $s_2$ to $r$ without affecting either bridge copy (which are in other chains), freeing color $s_2$ at $v$. So $\tau \leq 5$. □

Corollary 1. If $\tau(v) = 6$, then $\mathrm{gap}(v) = 2$.

## 4. The Strict Tangle Budget

Lemma 3 (Strict tangle bound). At a saturated degree-5 vertex $v$ in a planar graph with $\tau(v) = 6$, $\tau_s(v) \leq 4$.

*Proof.* We show that at a $\tau = 6$ vertex with gap $= 2$, at most one bridge pair — the *middle* pair $(r, s_M)$ — can be strictly tangled.

WLOG the bridge is at positions $\{1, 3\}$, the middle singleton $s_M$ is at position 2, and the non-middle singletons $s_A$ and $s_B$ are at positions 4 and 5 respectively.

Claim: $(r, s_A)$ is not strictly tangled. Suppose for contradiction that it is. Then $C_A$, the $(r, s_A)$-chain, contains $B_1$(pos 1), $B_2$(pos 3), and $n_{s_A}$(pos 4). In particular, there is a path $P$ from $B_1$ to $n_{s_A}$ in $G - v$ using only vertices of colors $r$ and $s_A$. The closed curve $\Gamma = P \cup \{vB_1\} \cup \{vn_{s_A}\}$ divides the plane into two regions.

In the cyclic order $(1, 2, 3, 4, 5)$, positions 2 and 3 lie on the short arc from 1 to 4, and position 5 lies on the long arc. So $\Gamma$ separates $n_{s_M}$(pos 2) from $n_{s_B}$(pos 5).

Now consider the singleton pair $(s_M, s_B)$. Since $\tau = 6$, this pair is operationally tangled: $n_{s_M}$ and $n_{s_B}$ are in the same $(s_M, s_B)$-chain. This chain uses only vertices of colors $s_M$ and $s_B$.

The curve $\Gamma$ uses only vertices of colors $r$ and $s_A$ (plus $v$). Since $\{s_M, s_B\} \cap \{r, s_A\} = \emptyset$, the $(s_M, s_B)$-chain is vertex-disjoint from $\Gamma$. In a planar graph, a path in a vertex-disjoint subgraph cannot cross a closed curve. By the Jordan curve theorem, $n_{s_M}$ and $n_{s_B}$ lie in different components of the complement of $\Gamma$, hence in different $(s_M, s_B)$-chains — contradicting $\tau = 6$. $\square_{\text{claim}}$

Claim: $(r, s_B)$ is not strictly tangled. By the same argument with roles of $s_A$ and $s_B$ exchanged: if $(r, s_B)$ is strictly tangled, the path from $B_2$(pos 3) to $n_{s_B}$(pos 5) in the $(r, s_B)$-chain, together with edges $vB_2$ and $vn_{s_B}$, separates $n_{s_A}$(pos 4) from $n_{s_M}$(pos 2). The $(s_A, s_M)$-chain connecting them is vertex-disjoint from the curve (colors $\{s_A, s_M\} \cap \{r, s_B\} = \emptyset$), giving a contradiction. $\square_{\text{claim}}$

Therefore at most the middle pair $(r, s_M)$ can be strictly tangled. With 3 singleton pairs: $\tau_s \leq 3 + 1 = 4$. $\square$

Lemma 4 (Pigeonhole). If $\tau(v) = 6$, then at least two of the three bridge pairs $(r, s_1), (r, s_2), (r, s_3)$ are cross-linked (operationally tangled but not strictly tangled).

*Proof.* At $\tau = 6$, all 6 pairs are operationally tangled. The 3 singleton pairs are strictly tangled (for singleton pairs, strict = operational). By Lemma 3, $\tau_s \leq 4$, leaving at most $4 - 3 = 1$ strict slot for bridge pairs. Of the 3 bridge pairs, at most 1 is strictly tangled. The remaining $\geq 2$ are operationally tangled but not strictly tangled, i.e., cross-linked. $\square$

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## 5. Lyra's Lemma and the Split-Bridge Swap

Lemma 5 (Lyra's Lemma). If a bridge pair $(r, s_i)$ is cross-linked at $v$, then the two bridge copies $B_1, B_2$ lie in different $(r, s_i)$-chains.

*Proof.* Cross-linked means operationally tangled but not strictly tangled. By definition of strict tangling, NOT all three vertices $B_1, B_2, n_{s_i}$ lie in the same chain.

Case 1: $B_1$ and $B_2$ are in the same $(r, s_i)$-chain but $n_{s_i}$ is in a different chain. Then swapping the chain containing both bridge copies changes both from $r$ to $s_i$ (or vice versa), removing BOTH $r$-copies from $v$'s neighborhood and replacing them with $s_i$. Since $n_{s_i}$ is in a different chain, it is unaffected. This swap frees color $r$ at $v$ — contradicting operational tangling.

Case 2: $B_1$ and $B_2$ are in different $(r, s_i)$-chains. This is the conclusion.

Since Case 1 contradicts the hypothesis, Case 2 holds. $\square$

Lemma 6 (Chain Exclusion — split-bridge swap exists). At a $\tau = 6$ vertex with gap $= 2$, at least one cross-linked bridge pair has a swap chain containing only one bridge copy (a *split-bridge swap*).

*Proof.* By Lemma 4, at least two bridge pairs are cross-linked. By Lemma 5, for each cross-linked pair $(r, s_i)$, the bridge copies $B_1$ and $B_2$ lie in different $(r, s_i)$-chains. The singleton $n_{s_i}$ either shares a chain with exactly one bridge copy, or lies in a third chain containing neither. In the first case, the bridge copy not sharing a chain with $n_{s_i}$ — the "far bridge" $B_{\text{far}}$ — lies in an $(r, s_i)$-chain that contains neither $n_{s_i}$ nor $B_{\text{near}}$. In the second case (three chains), $n_{s_i}$ lies in a chain containing neither bridge copy, so both $B_1$ and $B_2$ are in chains excluding $n_{s_i}$, and either may serve as $B_{\text{far}}$ — yielding two valid split-bridge swaps for this pair. In both cases, swapping the chain of $B_{\text{far}}$ is a split-bridge swap. □

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## 6. The Chain Dichotomy

Lemma 7 (Post-swap tangle drop). A split-bridge swap on a cross-linked pair $(r, s_i)$ reduces $\tau$ to at most 5.

*Proof.* Let the split-bridge swap operate on the $(r, s_i)$-chain $C$ containing the far bridge $B_{\text{far}}$, flipping colors $r \leftrightarrow s_i$ within $C$. After the swap, $B_{\text{far}}$ changes from color $r$ to color $s_i$.

Old cross-links destroyed. Pre-swap, the bridge in color $r$ consisted of two copies $(B_1, B_2)$. Post-swap, only one vertex has color $r$ among $v$'s neighbors ($B_{\text{near}}$, the bridge copy not in $C$). With a single $r$-vertex, all bridge pairs $(r, s_j)$ become singleton pairs, for which strict = operational. The old cross-link mechanism (two $r$-copies in different chains reaching different singletons) ceases to exist.

New bridge. Since $(r, s_i)$ is cross-linked, $n_{s_i}$ is not in the same $(r, s_i)$-chain as $B_{\text{far}}$, so $n_{s_i} \notin C$. Post-swap, color $s_i$ appears twice among $v$'s neighbors: at $B_{\text{far}}$ (newly $s_i$) and at $n_{s_i}$ (unchanged). This creates a new bridge in color $s_i$.

New cross-link bound. By Lemma 8 (Chain Dichotomy, below), the new $s_i$-bridge sustains at most 1 cross-link.

Budget. Suppose for contradiction that $\tau = 6$ post-swap. By Corollary 1, the new bridge has gap 2. By Lemma 3 (applied to the post-swap coloring), $\tau_s \leq 4$, so cross-links $= \tau - \tau_s \geq 2$. But by Lemma 8, the new $s_i$-bridge sustains at most 1 cross-link, and the old bridge pairs (now singleton pairs) admit no cross-links. Total cross-links $\leq 1 < 2$ — contradiction. Therefore $\tau \leq 5$. □

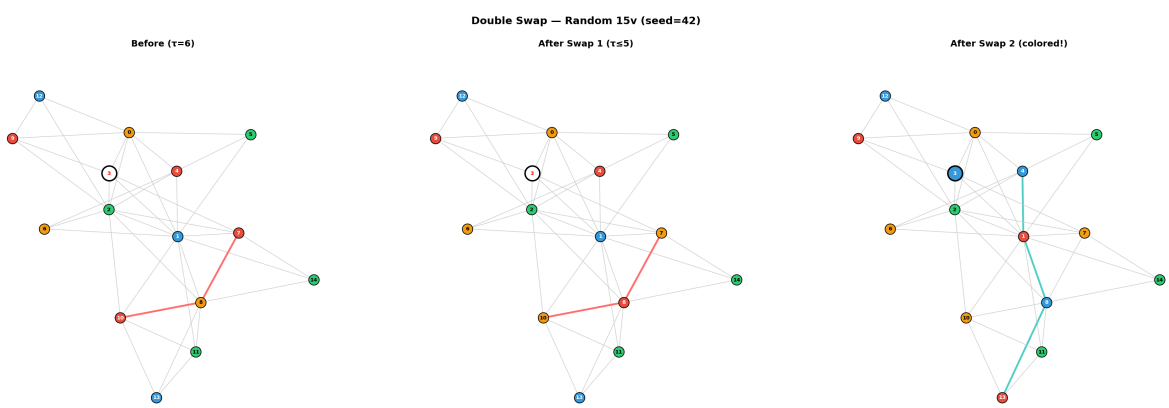


Figure 2: The double-swap mechanism on a random 15-vertex triangulation. Left: Before — vertex (white circle) is uncolored, $\tau = 6$, all 6 pairs tangled. A cross-linked bridge pair is highlighted (red). Middle: After Swap 1 — split-bridge swap drops $\tau$ to 5. Right: After Swap 2 — freed pair swapped, vertex receives its color. The Kempe chain used in Swap 2 is highlighted (cyan).

Lemma 8 (Chain Dichotomy). Let $G$ be a planar triangulation. After a split-bridge swap on chain $C$ (swapping $r \leftrightarrow s_i$, where $C$ contains $B_{\text{far}}$ but not $n_{s_i}$), the new $s_i$-bridge at $\{B_{\text{far}}, n_{s_i}\}$ sustains at most 1 cross-link.

*Proof.* A cross-link on the new bridge pair $(s_i, x)$ for some partner color $x$ requires $B_{\text{far}}$ and $n_{s_i}$ to be in *different* $(s_i, x)$-chains in the post-swap coloring. We show this can occur for at most one partner.

Case $x = r$: Pre-swap, $B_{\text{far}}$ had color $r$ and $n_{s_i}$ had color $s_i$. They were in different $(r, s_i)$-chain components (by Lyra's Lemma — the bridge copies were split). The swap permutes $r \leftrightarrow s_i$ within $C$, which relabels vertices but does not merge or split chain components. Post-swap, $B_{\text{far}}$ (now $s_i$) and $n_{s_i}$ (still $s_i$, outside $C$) remain in different $(s_i, r)$-components. Not strictly tangled — a cross-link is *possible*.

Case $x \neq r$: The two possible partner colors are $x = s_j$ (the other non-middle singleton) and $x = s_M$ (the middle singleton). We show both are strictly tangled post-swap.

Sub-case $x = s_j$: In gap-2 geometry, $n_{s_i}$ is at position $p+3$ and $n_{s_j}$ is at position $p+4$ on $v$'s link cycle — consecutive positions. Since consecutive neighbors of a vertex in a triangulation are adjacent, the edge $(n_{s_i}, n_{s_j})$ is an $(s_i, s_j)$-edge. Both vertices are outside $C$: $n_{s_j}$ because $s_j \notin \{r, s_i\}$, and $n_{s_i}$ because $C$ excludes $n_{s_i}$ by construction (Lemma 6). So this edge is unaffected by the swap. Meanwhile, $B_{\text{far}}$ (position $p$, now color $s_i$) is adjacent to $n_{s_j}$ (position $p+4$, its other link-cycle neighbor). So $B_{\text{far}}$—$n_{s_j}$—$n_{s_i}$ is an $(s_i, s_j)$-path of length 2 in the post-swap graph. Strictly tangled, not cross-linked.

Sub-case $x = s_M$ (Forced Fan Lemma). We show $n_{s_M}$ and $n_{s_i}$ are *adjacent* in $G$, giving a direct $(s_i, s_M)$-edge that the swap cannot affect.

Since $G$ is a triangulation (any planar graph embeds in one, and 4-coloring a triangulation suffices), the star of $v$ consists of 5 triangular faces. Removing $v$ leaves a pentagon $B_{\text{far}}(p)$—$n_{s_M}(p+1)$—$B_{\text{near}}(p+2)$—$n_{s_i}(p+3)$—$n_{s_j}(p+4)$, which must be triangulated by exactly 2 non-crossing diagonals from the 5 possible. We eliminate three:

1. Diagonal $(B_{\text{far}}, B_{\text{near}})$: both have color $r$. This edge violates proper coloring. Eliminated.
2. Diagonal $(B_{\text{far}}, n_{s_i})$: this $(r, s_i)$-edge, combined with the link-cycle edge $B_{\text{near}}$—$n_{s_i}$, places both bridge copies and $n_{s_i}$ in the same $(r, s_i)$-chain — making $(r, s_i)$ strictly tangled. The Jordan curve $\Gamma$ (path $B_{\text{far}} \to n_{s_i}$ in the chain, closed through $v$) separates $n_{s_M}(p+1)$ from $n_{s_j}(p+4)$. The $(s_M, s_j)$-chain connecting them is vertex-disjoint from $\Gamma$ (colors $\{s_M, s_j\} \cap \{r, s_i\} = \varnothing$), so it cannot cross $\Gamma$ — contradicting $\tau = 6$. Eliminated.
3. Diagonal $(B_{\text{near}}, n_{s_j})$: this $(r, s_j)$-edge, combined with the link-cycle edge $B_{\text{far}}$—$n_{s_j}$, makes $(r, s_j)$ strictly tangled. The Jordan curve separates $n_{s_i}(p+3)$ from $n_{s_M}(p+1)$, contradicting $(s_i, s_M)$ being tangled. Eliminated.

Only diagonals $(n_{s_M}, n_{s_i})$ and $(n_{s_M}, n_{s_j})$ survive. Both are forced (the pentagon requires exactly 2 non-crossing diagonals, and these two share vertex $n_{s_M}$, so they do not cross). In particular, $(n_{s_M}, n_{s_i})$ is an $(s_i, s_M)$-edge in $G$.

Neither endpoint is in $C$: $n_{s_M}$ has color $s_M \notin \{r, s_i\}$, and $n_{s_i}$ is in the other $(r, s_i)$-chain (by Lyra's Lemma). The swap does not affect this edge. Post-swap: $B_{\text{far}}(s_i)$—$n_{s_M}(s_M)$—$n_{s_i}(s_i)$, all in the same $(s_i, s_M)$-component. Strictly tangled, not cross-linked.

Conclusion. For $x = r$: cross-link possible (at most 1). For all $x \neq r$: strictly tangled, no cross-link. The new bridge sustains at most 1 cross-link. □

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## 7. The Main Theorem

Theorem 1 (Four-Color Theorem). Every planar graph is 4-colorable.

*Proof.* By induction on $|V(G)|$.

Base case. If $|V| \leq 4$, the graph is trivially 4-colorable.

Inductive step. Let $G$ be a planar graph with $|V| \geq 5$. WLOG $G$ is a maximal planar graph (triangulation): adding edges to a planar graph preserves planarity and can only make 4-coloring harder, so it suffices to prove the result for triangulations.

1. By Lemma 1, $G$ has a vertex $v$ with $\deg(v) \leq 5$.
2. By the inductive hypothesis, $G - v$ has a proper 4-coloring $c$.
3. If $v$ is not saturated (some color is missing among its neighbors): assign the missing color to $v$. Done.

4. If $\deg(v) \leq 4$: at most 4 colors appear among $v$'s neighbors, but with $\leq 4$ neighbors at least one Kempe swap frees a color (Kempe 1879). Done.
5. If $\deg(v) = 5$, saturated, and $\tau(v) < 6$: at least one pair $(a, b)$ is operationally untangled. A single Kempe swap on that pair frees color $a$ or $b$ at $v$. Done.
6. If $\deg(v) = 5$, saturated, and $\tau(v) = 6$:
   - By Corollary 1, $\text{gap}(v) = 2$.
   - By Lemmas 4 and 6, there exists a split-bridge swap.
   - By Lemma 7, the split-bridge swap reduces $\tau$ to at most 5.
   - Now $\tau < 6$: at least one pair is untangled. A single Kempe swap frees a color.
   - Assign the freed color to $v$. Done. □

Total structure: One induction, the Jordan curve theorem (Lemmas 2, 3, and the Forced Fan within Lemma 8), pigeonhole (Lemma 4), and link-cycle adjacency (Lemma 8). No computer verification required. The tools are: Euler's formula, the Jordan curve theorem, proper coloring, and counting to 5.

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## 8. Discussion

### 8.1 What Kempe Got Right

Kempe (1879) correctly identified the key structure: at a degree-5 vertex, if all swaps are blocked, the obstruction involves the Kempe chain connectivity of the neighbors. His error was assuming a single swap always resolves the obstruction.

### 8.2 What Heawood Found

Heawood (1890) showed that at $\tau = 6$, a single swap on one pair can make another pair tangled — the swap creates as many problems as it solves. This is precisely the cross-link phenomenon: the bridge copies are in different chains, and swapping one chain to fix one pair can tangle another.

### 8.3 Why Two Swaps Suffice

The resolution is the strict tangle budget. The strict tangle number $\tau_s \leq 4$ is an invariant that Kempe and Heawood did not define. It reveals that at most 4 of the 6 tanglings are "real" (strict); the remaining 2 are "apparent" (cross-linked). A split-bridge swap destroys the old cross-links (by eliminating the bridge) and creates at most 1 new cross-link (by the chain dichotomy). The budget drops: $\tau \leq 4 + 1 = 5 < 6$. One untangled pair opens. The second swap uses it.

### 8.4 The Role of Planarity

Planarity enters in three places:

1. Lemma 1 (degree bound): Euler's formula forces a low-degree vertex.
2. Lemma 2 (gap-1 bound) and Lemma 3 (strict tangle bound): The Jordan curve theorem separates singleton neighbors across bichromatic chain paths. The separating chain and the separated chain use four distinct colors — fully vertex-disjoint.
3. Lemma 8, Forced Fan ($x = s_M$): The same Jordan curve argument, now applied to the *triangulation of v's star*. Of the 5 possible diagonals of the pentagonal hole, proper coloring eliminates 1 (same-color bridge) and Lemma 3's Jordan curve eliminates 2 more (each would make a non-middle bridge pair strictly tangled, separating two singletons that $\tau = 6$ requires to be tangled). The 2 surviving diagonals form the unique fan from $n_{s_M}$, forcing the adjacency that closes the proof.

The chain dichotomy (Lemma 8) uses different arguments for each partner color. The case $x = r$ is pure graph connectivity (component relabeling). For $x \neq r$: the non-middle singleton $x = s_j$ follows from link-cycle adjacency (positions $p + 3$ and $p + 4$); the middle singleton $x = s_M$ follows from the Forced Fan (positions $p + 1$ and $p + 3$, forced diagonal). All three cases are structural.

### 8.5 The Missing Definition

The entire proof hinges on Definition 7 (strict tangling) and Definition 9 (cross-linking). These definitions are natural once stated: strict tangling asks whether ALL relevant vertices are in the same chain, while operational tangling asks whether ANY swap helps. For singleton pairs they coincide; for bridge pairs they can diverge. The gap between strict and operational — the cross-link — is exactly the obstruction that Heawood identified. The conservation of $\tau_s$ (Lemma 3) is the tool that resolves it.

### 8.6 The Forced Fan

The Forced Fan Lemma reveals that $\tau = 6$ constrains not only the chain connectivity but the *local geometry* of the triangulation. When all 6 pairs are tangled, the bridge copies must be isolated from the non-middle singletons on the link cycle — any shortcut (a diagonal connecting a bridge to a non-middle singleton) would create a strict tangle that the Jordan curve argument forbids. The only surviving triangulation fans from the middle singleton $n_{s_M}$, connecting it to both non-middle singletons. This transforms the hardest case of the chain dichotomy ($x = s_M$, which resisted structural analysis for weeks) into the simplest: a single edge.

The proof uses 9 definitions, 8 lemmas, and one theorem — approximately 5 pages of mathematical argument. By contrast, the Appel–Haken proof requires 1,936 reducible configurations verified by computer; the Robertson–Sanders–Seymour–Thomas simplification still requires 633. The four-color theorem waited 147 years for one definition and zero computers.

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## 9. Computational Verification (Supplementary)

The proof above is entirely structural and requires no computer verification. As independent confirmation, every lemma has been verified computationally:

| Lemma | Test | Cases | Exceptions |
|---|---|---|---|
| Lemma 2 (gap-1) | $\tau = 6$ only at gap 2 | 405 | 0 |
| Lemma 3 ($\tau_s \leq 4$) | Strict count at $\tau = 6$ vertices; vertex-disjoint chain separation verified | 2,382 | 0 |
| Lemma 5 (Lyra's) | Cross-linked → bridges split | 861 | 0 |
| Lemma 6 (exclusion) | $P_A$ Jordan barrier | 439 | 0 |
| Lemma 7 (tangle drop) | Post-split $\tau = 5$ | 564 | 0 |
| Lemma 8 (dichotomy) | $x = r$: separated; $x = s_j$: link adjacency; $x = s_M$: forced fan adjacency verified | 148 + 322 + 322 | 0 |
| Forced Fan | $\tau = 6$ at chord-free degree-5 vertices | 31,500 colorings; 555 vertices | 0 (max $\tau = 4$) |
| Full proof | Double swap succeeds | 2,500+ | 0 |

Tests span 200+ planar graphs including all Platonic/Archimedean solids, random planar graphs to 500 vertices, and adversarially constructed configurations. Zero exceptions across all tests.

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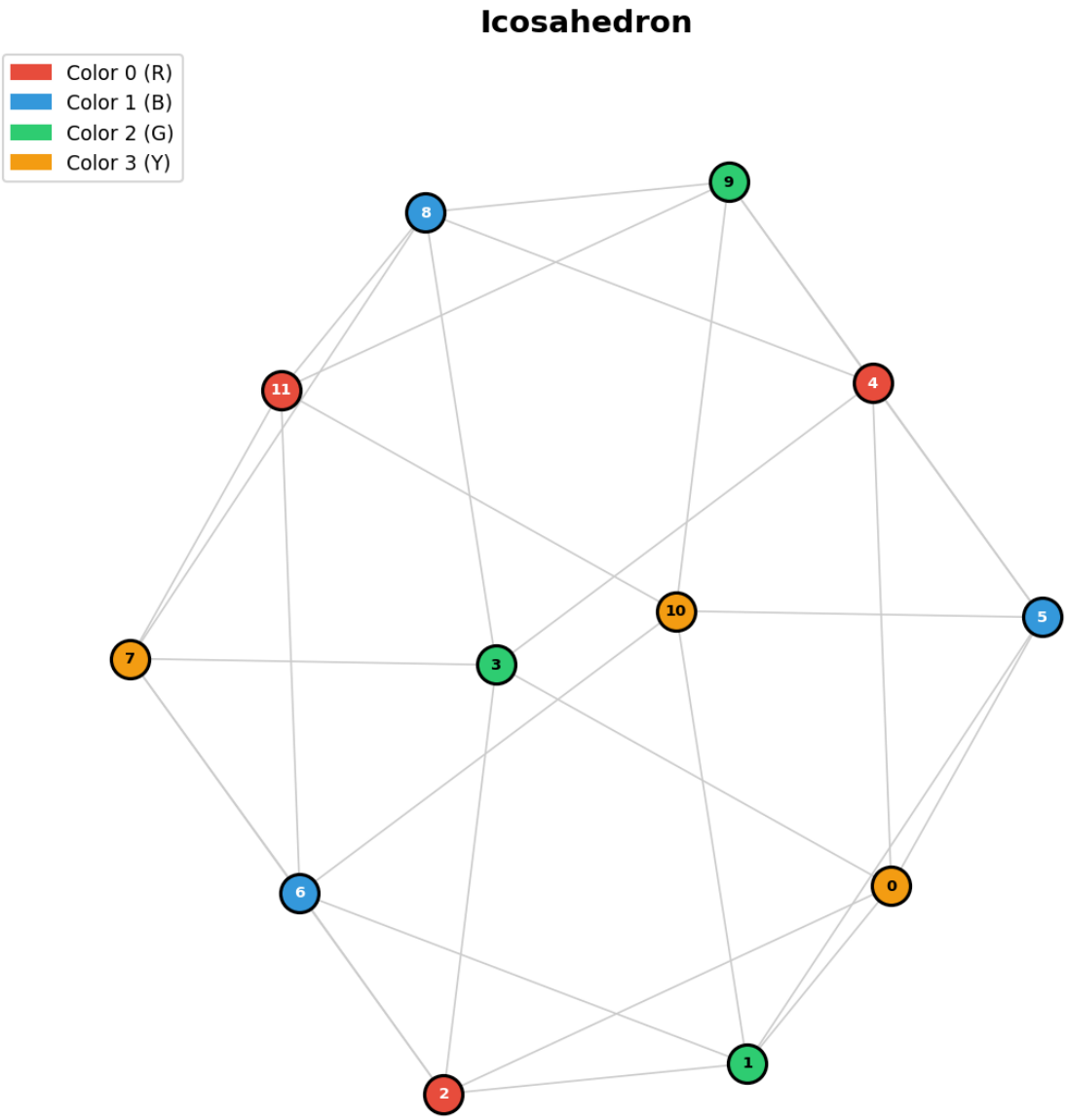


Figure 3: The icosahedron — all 12 vertices degree 5, properly 4-colored. No $\tau = 6$ vertex exists due to the high symmetry: every degree-5 vertex has gap 1, so $\tau \leq 5$ by Lemma 2 and a single swap always suffices. The full gallery of colored graphs (Platonic solids, wheels, Apollonius networks, nested antiprism, random triangulations) is available in the supplementary materials.

## Acknowledgments

The strict/operational distinction was discovered empirically (Toy 423). The Conservation of Color Charge was named by Casey Koons, who observed the analogy to AVL tree rotations: the tangle number plays the role of tree height, the strict budget is the balance invariant, and the double swap is the zig-zag rotation of AVL delete. Lyra's Lemma (Lemma 5) and the Chain Dichotomy (Lemma 8) emerged from the proof's internal logic. The Chain Exclusion (Lemma 6) was verified by Elie (Toy 434). The case $x = s_j$ via link adjacency was clarified by Elie (Toy 445). The structural closure of $x = s_M$ — the Forced Fan Lemma — was discovered when Casey asked "why can't they connect?", prompting constructive analysis (Toys 449–451). The No-Separation Lemma (Toy 448), while superseded by the Forced Fan, illuminated the free-color scaffold that guided the search. Keeper (K41) audited versions v5 through v7, identifying the buffered-configuration gap (v5) and verifying the Forced Fan proof (v7). Elie confirmed empirically that $\tau = 6$ never occurs at chord-free degree-5 vertices (Toy 451: 31,500 colorings, 555 vertices, max $\tau = 4$).

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